

Cisco CCNA 200-301 - The Complete Guide To Getting Certified

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Chapter 1

OSI Layer 3 - The network layer

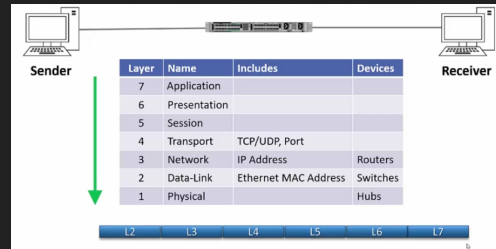
1.1 Introduction

- Format of the IP header.
- Format of IP addresses.
- Different types of traffic: unicast, broadcast, and multicast.
- Subnet mask.

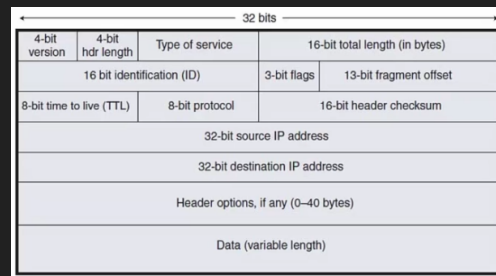
1.2 The IP header

- The network layer:
 - The network layer is responsible for routing packets to their destination and for Quality of Service.
 - * Quality of service: when some levels of traffic require better quality of traffic, so for example, a voice or video is sensitive to delay so it will need a better quality service than something like email.
 - IP (Internet Protocol) is the best known Layer 3 Protocol. IPv4 is the focus of this section.
 - * There is also IPv6 which is a newer protocol.
 - It is a connectionless protocol with no acknowledgements at Layer 3.
 - * You can have reliable traffic by using TCP at layer 4, but itself has no acknowledgement. You can also have reliable traffic by building it into the upper layers.
 - Other Layer 3 Protocols include ICMP (Internet Control Message Protocol) and IPSec.
 - * The ICMP is used for troubleshooting, IPSec is for secure and encrypted communications. There are others as well but these are the most common.
- IP addressing:
 - IP addressing is a logical addressing scheme which is implemented at layer 3.
 - The network designer uses IP addressing to partition the overall network into smaller 'subnets'.
 - This improves performance and security and makes troubleshooting easier.
 - * It improves performance because instead of having a big network we have a set of small 'subnets' and we can keep the traffic to the particular subnet that it needs to be on rather than everywhere.
 - * We also get better security. Example: if we have accounting servers, we can put it in it's own subnet and better control who has access to those servers.

- * Troubleshooting becomes easier because we now know that if a problem occurs it will show only on that particular subnet rather than in the entire network.
- Layer 2 MAC addresses use one big flat addressing scheme. There is no logical separation between networks at layer 2, it's done at Layer 3.
 - * IP addressing is a logical addressing scheme, whereas MAC addresses are a big flat scheme.
- OSI Reference Model - Encapsulation:



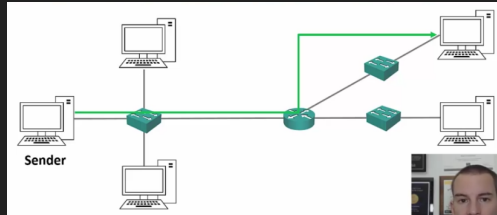
- The IP header:



- Version (4 bit): Ip version 4 or version 6.
- header length: (4 bit): the length of the header.
- Type of service: this is actually the quality of service information, we can mark the packet with quality of service information so that we can make routers respond to this information.
- total length (16 bit): length of the packet.
- The second row of information (identification, flags, fragment offset) handles the fragments. Fragments: packets have a maximum size, when the information that is to be sent exceeds the maximum packet size it is split up into several packets and sends them, these split up packets are called fragments.
- time to live: everytime a packet goes through a router, the router will decrement the TTL by one, if it goes down to 0 the router drops the packet. This is to avoid routing loops, routing loops are when a packet is being sent through the same routers endlessly without being able to find its destination because of some error, to prevent this the time to live is decreased to be able to discard the packet once it has reached the maximum amount of times. This doesn't fix the problem in the network that caused the routing loop, but it will allow us to mitigate a potentially growing problem in our network.
- 8-bit protocol: specify the layer 4 information type, usually TCP or UDP.
- Check sum: to ensure that the packet has not been corrupted in transit.
- source IP address (32 bit): from where the packet is coming from.
- destination IP address (32 bit): to where it is headed.
- Header options (0-40 bytes): this field is used to send any additional information in the header, it is not typically used.
- Data (variable length): the actual data being transmitted.

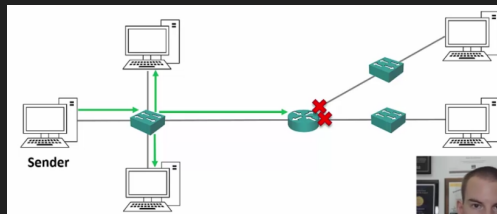
1.3 Unicast, Broadcast, and Multicast Traffic

- Unicast, Broadcast, and Multicast Traffic
 - There are 3 main IP traffic types: unicast, broadcast, and multicast.
 - Unicast traffic is to a single destination host.
 - Broadcast traffic is to all hosts on a subnet.
 - Multicast traffic is to multiple interested hosts.
- Unicast traffic:



- The traffic is coming from a sender, and is going just to one receiver.

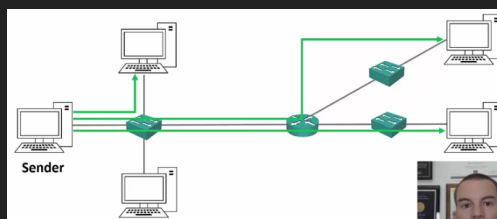
- Broadcast traffic:



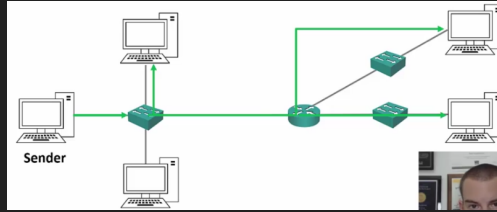
- The sender sends ONE copy of the traffic that will come into the switch, and then it gets to all of the hosts connected to the switch.
 - *Important: when a broadcast is sent in a subnetwork, routers do not forward broadcast traffic, only switches do that.

- Unicast Traffic to Multiple Hosts:

- With unicast you send out one individual copy of the requested data to the other hosts. This can potentially become expensive because of the extra bandwidth and can cause a lot of overhead.



- When you unicast a message to several computers you send to each individual destination host a different copy. With multicast you send one copy that goes to several different receivers, with this arrangement you don't produce a different copy for all of the interested receivers, you send out one, and that one copy gets forwarded to all the different receivers.



- Multicast goes to multiple receivers but only if the receivers request it, with broadcast all hosts on a network get the data regardless of request, multicast is targeted.
- *Analogy: multicast is like tuning to a radio station, you have to tune in to get that particular radio station.

1.4 Converting from decimal to binary

- Counting in decimal:

- Humans are conditioned to count in decimal.
- For each 'column' in a number we have 10 possible choices, from 0-9.
- Every time we add a column to the left, the value is multiplied by 10.
- We start with '1s' as the further right column.

236 is two 100's, three 10's, and six 1's.

1000's	100's	10's	1's	
0 (236)	2 (36)	3 (6)	6 (0)	236

- Writing 236 implies the following:

- Counting in binary:

- Computers work in binary.
- Electrical impulses are either off or on, so there's only two choices (0 or 1), unlike 10 in decimal (0-9).
- For each 'column' in a number we have 2 possible choices, 0 or 1.
- Everytime we add a column to the left, the value is multiplied by 2.
- In this case the columns are like so:

236 in binary is 11101100

256	128	64	32	16	8	4	2	1
0 (236)	1 (108)	1 (44)	1 (12)	0 (12)	1 (4)	1 (0)	0 (0)	0 (0)
0	1	1	1	0	1	1	0	0

- The number we are going to get in the end is 11101100. We do this like so. A final check is that at the end of the process you should have 0 left over.

1.5 IPv4 Addresses

- IPv4 Addresses:
 - An IPv4 address is 32 bits long.
 - It is written as 4 'octets' in dotted decimal format.
 - For example 192.168.10.15
 - Each octet is 8 bits long, 4 octets are 32 bits.
- Seeing your IP address:

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::49e8:5e2:b9d8:15b6%8  
IPv4 Address. . . . . : 192.168.1.9  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::1%8  
                             192.168.1.1
```

- If we were to type 'ipconfig' in cmd (windows) or 'ifconfig' in the terminal (linux and macos) we would get all sorts of information regarding the IP address.
 - IPv4 address is printed, the subnetmask is also printed, and the default gateway is printed. A default gateway is used when traffic is going to another host on another subnet, when the destination host is in the same subnet it can get there directly but if it is not it will go through a router and enter a different subnet, the default gateway in that case is the router to the other subnets and networks.
 - To see the default gateway on linux or macos enter 'ip route'.
 - To get your ip in Cisco OS, enter the enable prompt and type 'show ip interface brief'.
- Static and automatic addressing:
 - The IP address is usually set manually on servers, printers and network devices such as routers and switches. On desktop computers it is usually assigned automatically through the Dynamic Host Configuration Protocol (DHCP).
 - * The reason you want to get your IP is set automatically on desktops is because you would not want to set it up manually because you would need to see which IP address is not in use and then set it up, instead you are automatically assigned one. With servers and routers you can do it manually because they are fewer and they are going to almost always be the same IP address.
 - To understand how the logical separation between subnets works, you need to understand the IP address in binary.

1.6 Calculating and IPv4 Address in binary

- Each octet in the IP address has a value ranging from 0 to 255.
 - Since an octet has 8 bits, and its binary we know that it has a maximum of 255.

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

- Notice if we replace the 8 bits with 0's we get 0. If we replace everything with 1 we get $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$.
- Example: Let's convert that 192.168.10.15 address to binary, starting with the first octet of 192.

- Write out the binary columns on a piece of paper to do this: The final result is 11000000

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

• $192 - 128 = 64$
 • $64 - 64 = 0$

- The second octet of 168.

128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

• $168 - 128 = 40$
 • 40 – 64 doesn't go
 • $40 - 32 = 8$
 • 8 – 16 doesn't go
 • $8 - 8 = 0$
 • The second octet is 10101000 in binary
 • $128 + 32 + 8 = 168$

• The first half of the IP address in binary notation is 11000000.10101000

- And so on.
- The final answer should be:

192.168.10.15 = 11000000.10101000.00001010.00001111

• 192.168.10.15 = 11000000.10101000.00001010.00001111

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

128	64	32	16	8	4	2	1
0	0	0	0	1	0	1	0

128	64	32	16	8	4	2	1
0	0	0	0	1	1	1	1

- To set the boundary between logical network (subnet), the IP address is combined with the subnet mask.
- The subnet masked has not been explained, it will be explained in the next section.

1.7 The subnet mask

- The subnet mask:
 - The subnet mask can be visualized by using the command prompt. You can enter the appropriate command to show you your IP and you will see your IP, default gateway, and it's subnet mask.
 - A host can send traffic directly to another host on the same subnet via switches.
 - For host to send traffic to another host in a different subnet, it must be forwarded by a router.
 - * Routers are devices that link our different subnets together and route the traffic between them.
 - The host therefore sends to understand if the destination is on the same or a different subnet in order to know how to send it.
 - * If the destination is on the same subnet it can send it there directly and no router is required. If the destination host is not in the same subnet then we must send it to a router, this router is the default gateway.
 - * The way that the sender host knows if the destination host is on the same network is by simply comparing the sender's and reciever's IP address and subnet mask, if the mask is the same then just switches are required and no routing is done.

- The subnet mask is used for this.
- The subnet mask is also 32 bits long and can be written in dotted decimal or slash notation.
- Network and host portion:
 - A host's IP address is divided into a network portion and a host portion.
 - The subnet mask defines where the boundary is.
 - The easiest way to explain this is through example...
- Subnet 'Masking':
 - Let's say the host's IP address is 192.168.10.15 and its subnet mask is 255.255.255.0
 - We write the IP address out in binary notation, and then the subnet mask underneath.
 - Starting with IP=192.168.10.15 and subnet mask=255.255.255.0 we convert to binary:

192.168.10.15 = 11000000 10101000 00001010 00001111
 255.255.255.0 = 11111111 11111111 11111111 00000000

- * Notice the IP address, the IP address actually contains part of the subnet mask, we just need to know where it starts and where it ends. The way we know when it starts is by simply seeing the subnet mask, whenever there is a 1 aligned in the position, this means that part of the IP is also the network address.
- * So in this example notice that the first 24 bits in the subnet mask are set to 1. This means that the first 24 bits – or from bit 0-23 – of the IP are part of the network address.
- * It is counter intuitive but just see how the numbers are lined up. We convert them to binary and align them up vertically as for the first bit of the IP address to correspond with the first bit of the subnet mask, same with the second, third, ... all the way up to the 32nd bit.
- * Now you see the subnet mask, wherever there is a 1, that is the network address.

• 192.168.10.15 / 255.255.255.0

192	168	10	15	255	255	255	0
11000000	10101000	00001010	00001111	11111111	11111111	11111111	00000000

- * In the picture above, notice the red line, the 1s come up to the third octet. To the left of that border lies the network address, to the right of that border lies the host address.
- The IP address is compared ('masked') with the subnet mask.
- A '1' in the subnet mask indicates that bit in the IP address is part of the network address.
- A '0' indicates the bit is part of the host address.
- Below you can see the network portion:

192	168	10	15	255	255	255	0
11000000	10101000	00001010	00001111	11111111	11111111	11111111	00000000

- Below you can see the host portion:

192	168	10	15	255	255	255	0
11000000	10101000	00001010	00001111	11111111	11111111	11111111	00000000

- Local subnet or routed traffic:

• 192.168.10.15 / 255.255.255.0

192	168	10	15	255	255	255	0
11000000	10101000	00001010	00001111	11111111	11111111	11111111	00000000

- If the host wants to communicate with another host with an IP address which also begins with 192.168.10 (network portion for the example 192.168.10.20), it knows it's on the same subnet and it can send the traffic directly.
- If it wants to communicate with another host with any other network address (for example 192.168.11.20), it knows it has to send the traffic via a router.
- For a destination address to be in the same subnet, the network portion has to be exactly 192.168.10.
- Otherwise it's in a different subnet and traffic must be sent via a router.

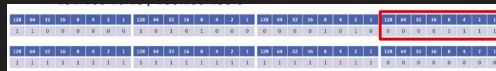
- Valid subnet masks:



- The subnet mask always begins with contiguous '1's.
- For example: 11111111.11110000.00000000.00000000 is a legal subnet mask.
- 11101101.11110000.00001111 is not.
- With IP addresses the '0's and '1's can be mixed, in the subnet mask it is always a '1's and then '0's, we don't mix them up in subnet masks.

- The host portion:

- The host portion of the address is available to be allocated to the different hosts on the subnet (eg PCs, Servers, Printers, Router interfaces and Switch Management Addresses).
- With two exceptions (that will be explained in a moment).
- Again, this is the host portion of the address:



- Host addresses:

- The host portion of the address specifies the individual host and must be unique on that subnet.
- Hosts do not have to be numbered sequentially.
 - * Any number can be assigned as long as they are not duplicated.
- If the network portion of the address is 10.10.10.10, you can have a host with IP address 10.10.10.10 and another with 10.10.10.20.
- You can't have two different hosts both with IP address 10.10.10.10. That would be a duplicate IP address. Whenever another host sent traffic to 10.10.10.10, the network wouldn't know which one to send it to.
- We have host 10.10.10.10 on one subnet and host 10.10.20.10 on another subnet.

- The network address (network ID) FIRST EXCEPTION:

- The first exception to the host portion is that all '0's in the host portion designates the network address and is not allowed to be allocated to a host.
 - * Basically you can pick the host portion to be anything except all 0s because all 0s it is assumed to be the network address.

- In our example the network address is 192.168.10.0. Remember the network address is also 32 bits, it just so happens that when you have the host portion in all 0s, that has been reserved for the network address and you cannot allocate it to a host. In our example the network address is 192.168.10.0.

192	168	10	0
1 1 0 0 0 0 0 0	1 0 1 1 1 1 1 0	1 0 1 1 0 1 0 1	0 0 0 0 0 0 0 0

- The broadcast address SECOND EXCEPTION:
 - The second exception to host numbers are all 1s.
 - All '1's designates the directed broadcast address for the subnet.
 - Traffic with this destination address will be sent to all hosts in the subnet.
 - In our example the broadcast address is 192.168.10.255.
 - So all 0s and all 1s in the host's portion are the exceptions.

192	168	10	255
1 1 0 0 0 0 0 0	1 0 1 1 1 1 1 0	1 0 1 1 0 1 0 1	1 1 1 1 1 1 1 1

- Host addresses:
 - So taking out the exceptions of the host portion (all 1's and all 0's) that leaves the host portion free to be anything from 1-254. In our example that leaves 192.168.10.1 to 192.168.10.254 available to be allocated to hosts. All hosts connected to network address 192.168.10.0 can send traffic directly through switches because they are in the same subnet without going via their default gateway router.

1.8 Slash notation

- Subnet mask in slash notation:
 - So notice that the subnet mask is always contiguous 1s followed by contiguous 0s. This is long and tedious, there is a better way of writing it down. For IP address 192.168.10.15 and subnet mask 255.255.255.0 we can simply enumerate the 32 bits of the subnet mask and just write down the range from 1-n where the '1's are at.
 - Because the subnet mask always begins with contiguous '1's, it will be 1 to 32 bits long counting from left to right.
 - This allows us to write the subnet mask in slash notation which is more convenient than dotted decimal for network diagrams or in conversation.
 - So look at the following example:

192.168.10.15 / 255.255.255.0			
128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1
1 1 0 0 0 0 0 0	1 0 1 1 1 1 1 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 1 1
128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1	128 64 32 16 8 4 2 1
1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0

- We see that the first three octets are 1s and the last octet is all 0s. So the slash notation for this case would be /24 because the first 24 bits are set to 1, the remaining are 0s.
- In cisco IOS the subnet mask is written in dotted decimal.

- Subnet mask in slash notation example 2:



- This example can be written as either 10.10.10.15 255.0.0.0 or 10.10.10.15/8.
- The network address is 10.0.0.0/8.
- The last three octets is the host portion.
- The available addresses would be 10.0.0.1 to 10.255.255.254, not 10.255.255.255 because that is the broadcast address.